# Homework Week2

## Sequence Alignment Fundamentals

http://thegrantlab.org

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This unit's homework consists of both (1) an online knowledge assessment quiz (see online) and (2) a Needleman-Wunsch dynamic programming assessment exercise (this document). Both components contribute 50% to this unit's grade. For the later we have two sample sequences, and we'd like to use the Needleman-Wunsch algorithm discussed in class to align them.

| | | T | A | T | A | G | C |
|---|---|---|---|---|---|---|---|
| | 0 | -2 | -4 | - 6 | -8 | -10 | -12 |
| G | - 2 | -1 | -3 | - 5 | -7 | -6 | -8 |
| T | - 4 | 0 | -2 | - 1 | -3 | -5 | -7 |
| T | - 6 | -2 | -1 | 0 | -2 | -4 | -6 |
| A | - 8 | -4 | 0 | - 2 | 2 | 0 | -2 |
| T | -10 | -6 | -2 | 2 | 0 | 1 | -1 |
| C | -12 | -8 | -4 | 0 | 1 | -1 | 3 |

Sequence 1: **TATAGC**

Sequence 2: **GTTATC**

Using a **match score of +2**, a **mismatch score of -1**, and a **gap score of -2**. Fill in the table and translate it into a alignment.

Please submit your completed answer via **gradescope** . This should be titled "*02. Global Alignment HW Week2*". You can submit this document as a PDFor a photo of a separate page with your completed **alignment matrix** along with your **aligned sequences** and their **optimal score**.

Alignment 1:
TATAGC
GTTATC
optimal score = 3

Alignment 2:
-TATAGC
GT-TATC
optimal score = 3

| Step | Scoring Rubric/Assessment Criteria | Points | |
|---|---|---|---|
| 1 | Setup labeled alignment matrix | 1 | |
| 2 | Include initial column and row for GAPs | 1 | |
| 3 | All alignment matrix elements filled in | 1 | |
| 4 | Evidence for correct use of scoring scheme | 1 | |
| 5 | Direction arrows drawn between all cells | 1 | |
| 6 | Evidence of multiple arrows to a given cell if appropriate | 1 | D |
| 7 | Correct optimal score position in matrix used | 1 | C |
| 8 | Correct optimal score obtained for given scoring scheme | 1 | B |
| 9 | Traceback path(s) clearly highlighted | 1 | A |
| 10 | Correct alignment(s) yielding optimal score listed | 1 | A+ |

(10 Total points)